

# Nicholas Richardson

## Curriculum Vitae

Institute of Applied Mathematics  
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[njericha.github.io](https://github.com/njericha)

### Research Interests

Audio/Signal Processing, Tensor Factorization, and Continuous Optimization

### Education

- Sep. 2022 – **Doctor of Philosophy**, *University of British Columbia*, Vancouver, BC  
(Jul. 2026) Institute of Applied Mathematics, supervised by Michael P. Friedlander  
Thesis: *Computational Techniques for Analysis of Mixed Signals*
- Sep. 2020 – **Master of Mathematics**, *University of Waterloo*, Waterloo, ON  
May 2022 Applied Mathematics Department, supervised by Giang Tran  
Thesis: *A Sparse Random Feature Model for Signal Decomposition*
- Sep. 2016 – **Bachelor of Mathematics**, *University of Waterloo*, Waterloo, ON  
Apr. 2020 Mathematical Physics Major, Pure Mathematics Minor

### Publications

- 2025 Joel Saylor, **Nicholas Richardson**, Naomi Graham, Robert Lee, and Michael P. Friedlander. Tracking cu-fertile sediment sources via multivariate petrochronological mixture modelling of detrital zircons. *Journal of Geophysical Research: Earth Surface*, 130, e2025JF008406.
- 2025 Naomi Graham, **Nicholas Richardson**, Michael P. Friedlander, and Joel Saylor. Tracing Sedimentary Origins in Multivariate Geochronology via Constrained Tensor Factorization. *Mathematical Geosciences*, 57, 601–628.
- 2023 Michael L. Waite and **Nicholas Richardson**. Potential Vorticity Generation in Breaking Gravity Waves. *Atmosphere*, 14(5):881.
- 2023 **Nicholas Richardson**, Hayden Schaeffer, and Giang Tran. SRMD: Sparse Random Mode Decomposition. *Communications on Applied Mathematics and Computation*, 6, 879–906.
- 2023 Lam Si Tung Ho, **Nicholas Richardson**, and Giang Tran. Adaptive group Lasso neural network models for functions of few variables and time-dependent data. *Sampling Theory, Signal Processing, and Data Analysis*, 21(2):28.

#### Submitted

**Nicholas Richardson**, Noah Marusenko, and Michael P. Friedlander.  
Multiple Scale Methods for Optimization of Discretized Continuous Functions.  
[arxiv.org/abs/2512.13993](https://arxiv.org/abs/2512.13993).

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## Teaching

- Sep. 2023 – **Instructor**, *University of British Columbia*, Vancouver, BC
- Apr. 2026
- Intro to Computing: Python (Fall 2025, Winter 2026)
  - Calculus I (Fall 2023, Winter 2025)
- Sep. 2022 – **Teaching Assistant**, *University of British Columbia*, Vancouver, BC
- Dec. 2024
- Calculus I (Fall 2024)
  - Calculus II (Winter 2023)
  - Calculus III (Fall 2022)
- Sep. 2022 – **Mathematics Learning Centre Tutor**, *University of British Columbia*, Vancouver, BC
- Apr. 2024
- June 2020 – **Mathematics & Physics Tutor**, *Self-Employed*
- Dec. 2024
- High School and Undergraduate Levels
- Jan. 2018 – **Teaching Assistant**, *University of Waterloo*, Waterloo, ON
- Dec. 2021
- Mathematics of Music (Fall 2020, 2021)
  - Special Topics in Mathematical Connections: Math & Music (Fall 2021)
  - Complex Analysis (Winter 2020)
  - Calculus 1 & 2 (Fall 2019, Winter 2018, 2020, 2021)

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## Awards

### McMaster University

- Sep. 2026
- Britton Postdoctoral Fellowship

### Society of Industrial and Applied Mathematics

- Jun. 2026
- OP26 Student Travel Award
- Jul. 2024
- AN24 Student Travel Award

### Natural Sciences and Engineering Research Council of Canada

- May 2023
- Canadian Graduate Scholarship – Doctoral (330 awards nationwide)
- May 2021
- Canadian Graduate Scholarship – Master's (840 awards nationwide)
- May 2019
- Undergraduate Student Research Award (3 149 awards nationwide)

### University of British Columbia

- May 2022
- Faculty of Science PhD Tuition Award
- Sep. 2022
- Four Year Doctoral Fellowship
- Sep. 2022
- President's Academic Excellence Initiative PhD Award

### University of Waterloo

- May 2021
- President's Graduate Scholarship
- Aug. 2020
- K.D. Fryer Gold Medal
- June 2020
- Dean's Honours List With Distinction
- May 2019
- President's Research Award
- Jan. 2019
- Paul Berg Memorial Award

## (Awards Continued)

- Jan. 2019 Arthur Beaumont Memorial Scholarship
- Sep. 2016 President's Scholarship of Distinction
- Sep. 2016 Alumni @ Microsoft Entrance Scholarship in Mathematics

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## Presentations & Posters

- (Jun 2026) **SIAM Conference on Optimization, Edinburgh, UK**  
Multiscale Methods for Optimization of Discretized Continuous Functions (Oral).
- Jan 2026 **SFU Operations Research Seminar, Burnaby, BC**  
Optimization and Applications of Unsupervised Signal Demixing (Oral).
- Dec 2025 **Geology Industry Training with Teck and BHP, Vancouver, BC**  
Tucker-1 Demixing Model: Under the Hood (Oral).
- July 2025 **Joint SIAM/CAIMS Annual Meetings, Montréal, QC**  
BlockTensorDecomposition.jl: An Optimization Framework for Constrained Tensor Factorization (Oral).
- July 2025 **CAIDA/TrustML Workshop — ICML Visitors @ UBC, Vancouver, BC**  
Unsupervised Signal Demixing with Tensor Factorization (Poster).
- July 2025 **Vision and Learning Workshop @ ICML, Vancouver, BC**  
Unsupervised Signal Demixing with Tensor Factorization (Poster).
- Dec. 2024 **CMS Winter Meeting, Richmond, BC**  
Density Separation with Tensor Factorization (Oral).
- Oct. 2024 **INFORMS Annual Meeting, Seattle, WA**  
Density Separation with Tensor Factorization (Oral).
- Sep. 2024 **West Coast Optimization Meeting, Vancouver, BC**  
Density Separation with Tensor Factorization (Oral).
- July 2024 **SIAM Annual Meeting, Spokane, WA**  
Density Separation with Tensor Factorization (Oral).
- Oct. 2023 **Biennial Meeting of SIAM Pacific Northwest Section, Bellingham, WA**  
Non-negative Tensor Decompositions for Unsupervised Learning (Oral).
- June 2021 **Canadian Applied and Industrial Mathematics Society, Waterloo, ON**  
Sparse Random Wavelet Representation & Decomposition (Poster).  
Runner up in annual poster competition

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## Software

### Published Code & Packages

- 2026 UBC Thesis Typst Template, *Typst*  
[github.com/njericha/ubc-thesis-typst-template](https://github.com/njericha/ubc-thesis-typst-template)
- 2024–2025 MatrixTensorFactor.jl & BlockTensorFactorization.jl, *Julia*  
[github.com/MPF-Optimization-Laboratory/BlockTensorFactorization.jl](https://github.com/MPF-Optimization-Laboratory/BlockTensorFactorization.jl)
- 2023 SedimentSourceAnalysis.jl, *Julia*  
[github.com/njericha/SedimentSourceAnalysis.jl](https://github.com/njericha/SedimentSourceAnalysis.jl)

(Software Continued)

- 2022 Sparse Random Mode Decomposition, *Python*  
[github.com/GiangTTran/SparseRandomModeDecomposition](https://github.com/GiangTTran/SparseRandomModeDecomposition)
- 2017 Drasil, *Haskell*  
[jacquescurette.github.io/Drasil](https://jacquescurette.github.io/Drasil)

Computer Skills

Julia, Python, MATLAB, Quarto, Typst,  $\LaTeX$ , Adobe Software, Haskell

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## Journal Papers Reviewed

- 2025 SIAM Journal on Scientific Computing (1 review).
- 2023–2026 Springer’s Journal of Optimization Theory and Applications (3 reviews).

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## Community & Leadership

- Sep. 2025 – **SIAM Student Chapter**, University of British Columbia  
(Jul. 2026) ○ Co-organizer for the first UBC Applied Mathematics Meeting (Oct. 2025)
- Sep. 2023 – **Math Graduate Committee**, University of British Columbia  
(Jul. 2026) American Mathematical Society Graduate Student Chapter  
○ Co-organizer and Seminar Host (2024–Present)  
○ Social Convenor (2023–Present)

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## Student Supervision

- May. 2026 – **Michael Yang**, *Incoming Master’s Student Summer Project*,  
(Aug. 2026) University of British Columbia  
Moment Matching for Density Estimation.
- May. 2026 – **Mehdi Naami**, *Undergrad Student Research Project*,  
(Aug. 2026) University of British Columbia  
Block Descent and Factorization with Automatic Differentiation.
- Sep. 2024 – **Noah Marusenko**, *Undergrad Student Research Project*,  
Jan. 2025 University of British Columbia  
Multiple Scale Methods for Tensor Factorizations.

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## Additional Research

- May 2019 – **Computational Fluids Research Assistant**, *University of Waterloo*,  
Aug. 2019 Waterloo, ON, supervised by Michael L. Waite
- May 2017 – **Software Engineering Research Assistant**, *McMaster University*,  
Aug. 2017 Hamilton, ON, supervised by Spencer Smith & Daniel Szymczak